

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I BALA S(192524427) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: bala

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1.Type-1 and Type-2 hypervisors:

Feature	Type-1 Hypervisor (Bare-Metal)	Type-2 Hypervisor (Hosted)
Installation	Runs directly on the physical hardware	Runs on top of a host operating system
Performance	Very high performance with low overhead	Lower performance due to host OS overhead
Security	More secure because there is no host OS attack surface	Less secure since vulnerabilities in the host OS can affect the hypervisor
Resource Management	Efficient CPU, memory, and storage allocation	Less efficient because resources are shared with the host OS
Reliability	Highly reliable and suitable for continuous operation	Depends on the stability of the host operating system
Scalability	Easily supports hundreds or thousands of virtual machines	Limited scalability for enterprise workloads
Examples	VMware ESXi, Microsoft Hyper-V, Xen, KVM	VMware Workstation, Oracle VirtualBox, Parallels Desktop
Common Use	Enterprise cloud data centers and production environments	Software testing, learning, and desktop virtualization

1. Performance

- Type-1 Hypervisor: Directly accesses hardware, resulting in minimal latency, better CPU utilization, and faster virtual machine performance. It is ideal for mission-critical applications that require high availability.
- Type-2 Hypervisor: Since it relies on the host operating system, additional processing overhead reduces performance, making it unsuitable for heavy enterprise workloads.

2. Security

- Type-1 Hypervisor: Offers stronger isolation between virtual machines and fewer attack points because there is no underlying host OS.
- Type-2 Hypervisor: The host operating system can become a security weakness. If the host OS is compromised, all virtual machines may be at risk.

3. Manageability

- Type-1 Hypervisor: Provides centralized management, live migration, high availability, automated resource allocation, and disaster recovery features. These capabilities simplify managing large cloud environments.
- Type-2 Hypervisor: Easier to install and use but lacks advanced enterprise management features. It is better suited for personal use or development environments.

2.Virtualization Architecture for Workload Isolation in a Cloud Datacenter:

- A cloud provider can securely host finance, healthcare, and student portal workloads on the same physical server by using virtualization architecture.
- A hypervisor creates multiple isolated Virtual Machines (VMs), each with its own operating system, applications, memory, CPU allocation, storage, and virtual network.
- Although the VMs share the same hardware, they operate independently, preventing one workload from directly accessing another.

How Virtualization Supports Isolation:

1. Virtual Machine Isolation

- Each workload runs in its own VM with a separate operating system.
- A failure or compromise in one VM does not directly affect other VMs.

2. Resource Isolation

- The hypervisor allocates dedicated CPU, memory, storage, and network resources.
- Prevents one workload from consuming excessive resources and impacting others.

3. Network Isolation

- Virtual LANs (VLANs), virtual switches, and software-defined networking (SDN) separate network traffic.
- Finance, healthcare, and student portal data remain isolated during communication.

4. Storage Isolation

- Each VM uses separate virtual disks and access permissions.
- Sensitive data from one department cannot be accessed by another VM without authorization.

5. Access Control

- Role-Based Access Control (RBAC), authentication, and permissions ensure only authorized users and administrators can access specific workloads.

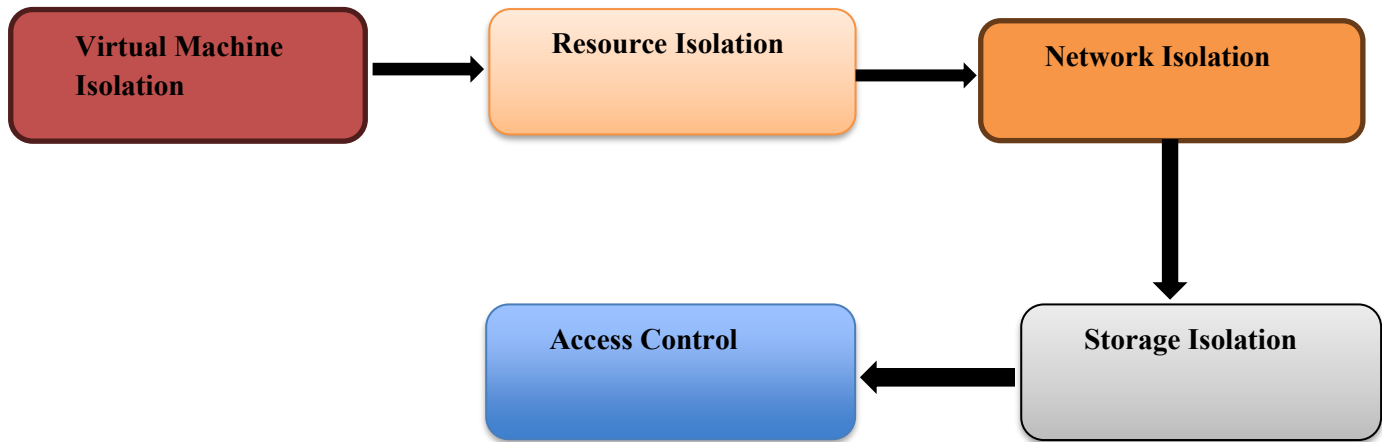
Major Attack Surfaces

1. Hypervisor Attacks

- The hypervisor controls all virtual machines.
- If an attacker exploits a vulnerability in the hypervisor, they may gain control over multiple VMs hosted on the same server.
- Impact: Unauthorized access, data theft, or disruption of finance, healthcare, and student portal services.
- Mitigation: Regular security updates, secure hypervisor configuration, and minimal attack surface.

2. VM Escape

- VM escape occurs when malicious code inside a virtual machine breaks out of the VM and interacts with the hypervisor or other VMs.
- Impact: Cross-VM attacks, data leakage, and compromise of multiple workloads.
- Mitigation: Strong VM isolation, timely patching, hardware-assisted virtualization, and continuous monitoring.



3. Analysis of VM Host Utilization Snapshot:

Metric	Observed Value	Status
CPU Utilization	92%	Very High
Memory Utilization	88%	High
Storage I/O Latency	35 ms	High
Average VM Boot Time	170 s	Slow

1. CPU Bottleneck

- Observation: CPU utilization is 92%, indicating the host processors are nearly saturated.
- Impact:
 - Slower execution of virtual machines.
 - Increased response time for applications.

- Delays in scheduling VM processes.

2. Memory Bottleneck

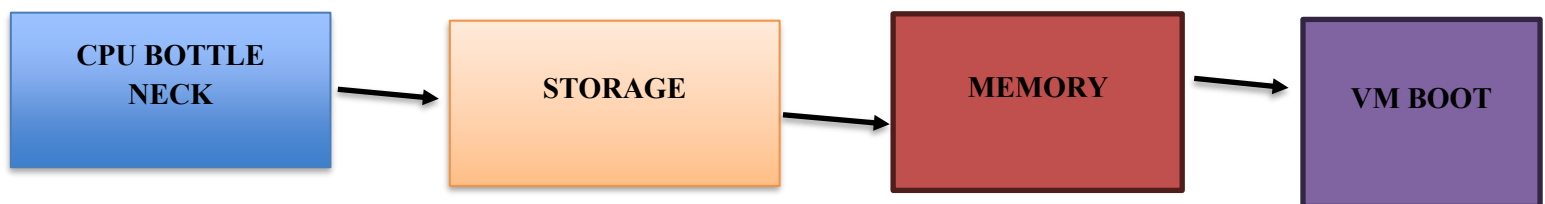
- Observation: Memory utilization is 88%, leaving very little free RAM.
- Impact:
 - Possible memory swapping or ballooning.
 - Reduced VM performance.
 - Increased application response times.

3. Storage I/O Bottleneck

- Observation: Storage latency of 35 ms is much higher than the typical target of less than 10 ms for virtualized environments.
- Impact:
 - Slow VM startup.
 - Delayed disk read/write operations.
 - Poor database and application performance.

4. Slow VM Boot Time

- Observation: Average VM boot time is 170 seconds.
- Likely Causes:
 - High CPU utilization delaying initialization.
 - Limited available memory.
 - High storage latency slowing operating system loading.
 - Simultaneous VM startups causing resource contention.



4. Software Defined Datacenter (SDDC) vs Traditional Datacenter:

- A Software Defined Data center (SDDC) is a modern data center architecture in which compute, storage, and networking are virtualized and managed through software.
- Unlike a traditional data center that relies on manual hardware configuration, an SDDC automates resource provisioning, management, and scaling, resulting in greater agility and efficiency.

Feature	Traditional Data center	Software Defined Data center (SDDC)
Resource Management	Manual hardware configuration	Software-based automated management
Compute	Dedicated physical servers	Virtual machines and containers
Storage	Physical storage devices	Software-defined storage (SDS)
Networking	Hardware switches and routers	Software-defined networking (SDN)
Provisioning Time	Hours or days	Minutes
Scalability	Limited and manual	Automatic and highly scalable
Flexibility	Low	High
Cost Efficiency	Higher operational costs	Better resource utilization and lower costs

1. Compute Virtualization

- Physical servers are divided into multiple Virtual Machines (VMs) using a hypervisor.
- Applications can be deployed, migrated, or scaled without changing physical hardware.
- Live migration allows workloads to move between hosts with minimal downtime.

Benefit: Faster deployment, better resource utilization, and improved availability.

2. Storage Virtualization

- Multiple physical storage devices are combined into a single virtual storage pool.
- Storage can be allocated or expanded dynamically according to application needs.
- Supports automated backup, replication, and disaster recovery.

Benefit: Flexible storage management, improved reliability, and simplified administration.

3. Network Virtualization

- Physical networking is abstracted using Software-Defined Networking (SDN).
- Virtual switches, routers, firewalls, and load balancers are configured through software.
- Network policies and security rules can be applied automatically.

Benefit: Rapid network provisioning, better security, and easier workload mobility.

Advantages of SDDC

- Rapid provisioning of IT resources.

- Automated management reduces manual effort.
- Efficient utilization of compute, storage, and network resources.
- Improved scalability to meet changing business demands.
- Faster disaster recovery and business continuity.
- Lower operational and maintenance costs.
- Centralized monitoring and management.

5. Identity Mismatch in a Multi-Cloud Federation:

- In a multi-cloud federation, different cloud providers (such as AWS, Azure, and Google Cloud) work together to provide services.
- Identity mismatches occur when each provider uses different identity systems, authentication methods, or user attributes, causing users to fail authentication or receive incorrect access permissions.

1. Different Identity Providers (IdPs)

- Each cloud provider may maintain its own user database and authentication mechanism.
- User identities are not synchronized across providers.

2. Inconsistent User Attributes

- One provider may identify a user by email, while another uses employee ID or username.
- Attribute mismatches lead to authorization failures.

3. Different Access Control Policies

- Each cloud provider may define roles and permissions differently.
- Users may receive excessive privileges or be denied access.

4. Privacy and Compliance Challenges

- User credentials and personal information may be duplicated across clouds.
- This increases the risk of data leakage and non-compliance with privacy regulations.

Secure Identity and Privacy Design Using Federation Concepts

Secure Identity and Privacy Design Using Federation Concepts

1. Federated Identity Management (FIM)

- Use a central Identity Provider (IdP) that authenticates users once.
- All cloud providers trust the central IdP instead of maintaining separate user identities.

- This enables Single Sign-On (SSO) across multiple cloud platforms.

Benefit: Users log in once and securely access services across all participating clouds.

2. Standard Federation Protocols

Use industry-standard authentication protocols such as:

- SAML (Security Assertion Markup Language) for enterprise web applications.
- OAuth 2.0 for secure authorization.
- OpenID Connect (OIDC) for user authentication.

Benefit: Ensures interoperability between different cloud providers.

3. Identity Federation

- Establish trust relationships between cloud providers.
- Providers accept identity assertions issued by the trusted IdP instead of creating separate accounts.

Benefit: Eliminates identity mismatches and simplifies user management.

4. Role-Based Access Control (RBAC)

- Define common roles such as Admin, Finance User, Healthcare User, and Student User.
- Map these roles consistently across all cloud providers.

Benefit: Ensures users receive the correct permissions regardless of the cloud platform.

5. Privacy Protection

- Share only the minimum user attributes required (principle of least privilege).
- Encrypt identity tokens during transmission.
- Use Multi-Factor Authentication (MFA) to strengthen authentication.
- Maintain audit logs for all authentication and authorization activities.

Benefit: Protects sensitive user information while meeting security and privacy requirements.

Benefits of the Proposed Design:

- Single Sign-On (SSO) across multiple cloud providers.
- Consistent user identity and authorization.
- Reduced administrative overhead.
- Improved security through centralized authentication.
- Better privacy by minimizing data sharing.
- Easier compliance with regulations such as GDPR and HIPAA.

